Examiner only

3. *(a)* State the principle of conservation of momentum. [2]

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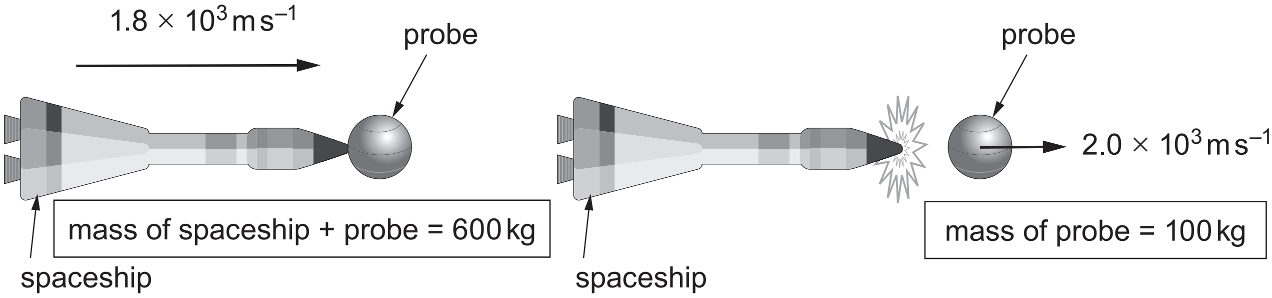
(b) A space probe is attached to a large spaceship as shown. The probe can be ejected from the space ship by an explosion. The spaceship and probe together have a mass of 600 kg and they travel in a straight line through deep space at $1.8 \times 10^3\,\mathrm{m\,s^{-1}}$. Explosives are detonated, separating the probe from the spaceship. Immediately after the explosion the probe, of mass 100 kg, continues in the original straight line at $2.0 \times 10^3\,\mathrm{m\,s^{-1}}$.



before explosion **after explosion**

(i) Calculate the velocity of the spaceship immediately after the explosion. [2]

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(ii) Show that the kinetic energy possessed by the probe and spaceship after separation is **greater** than the kinetic energy they possessed before separation. [3]

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(iii) Account for this increase in kinetic energy. [1]

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(c) During the explosion, a mean force, F, acts on the probe for 2.0 ms. Calculate the value of F. [2]

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